

SAILAKSHMI B T

+91 91641 32885 | sailakshmibt2006@gmail.com | Chennai, India | linkedin.com/in/sailakshmi-65a696327 |
github.com/Saibts

PROFESSIONAL SUMMARY

Robotics and Automation undergraduate (CGPA 9.10/10.0) with hands-on experience in embedded systems, robotics, and computer vision. Skilled in Arduino-based prototyping, sensor interfacing, and AI-driven applications. Seeking internship opportunities in robotics, embedded systems, or AI/computer vision.

EDUCATION

B.E., Robotics and Automation, Madras Institute of Technology (CGPA: 9.10/10.0) 2024 - 2028

Higher Secondary Certificate (2024): 94.16% | SSC (2022): 95.2% | JEE (Main) 2024: 81.82 percentile

INTERSHIPS & TRAINING

Computer Vision and AI Intern - PixelBotix June 2026

- Completed 4-week training in image processing and object detection, building and deploying 3 computer vision and AI projects.

Industrial Robotics and Automation Training - Birla Carbon and KUKA Feb 28, 2026

- Trained in industrial robotic arm programming, gaining exposure to Industry 4.0 concepts and KUKA safety protocols.

PROJECTS

AgileX Scout Mini ROS 2 Jazzy Integration ROS 2 Jazzy, C++, Python, Gazebo, RViz2

- Developed ROS 2 Jazzy driver nodes for real-time telemetry and odometry tracking, validated across multiple Gazebo/RViz2 simulation scenarios.

Student Life Concierge AI Agent Google ADK, Gemini API, FastAPI, Python, Pytest

- Built a multi-agent FastAPI backend with prompt-injection defenses to automate student scheduling, using Google ADK and Gemini API as part of the Kaggle and Google 5-Day AI Agents Intensive.

Single-Axis Gimbal Stabilization System Arduino Nano, MPU-6050, SG90 Servo

- Engineered a closed-loop PID gimbal stabilizer using an MPU-6050 IMU on Arduino Nano, achieving sub-degree stabilization accuracy.

DESIGN & PATENTS

Mechanical Rotary Device Design No. 476802-001 | SolidWorks, Design Patent Pending

- Designed a dual-mode Geneva wheel and dog clutch rotary mechanism in SolidWorks, combining two motion-control functions into one assembly; filed for design patent.

HACKATHONS & COMPETITIONS

Smart India Hackathon 2025 - Finalist Delhi, 2025

- Collaborated on reverse-engineering multi-architecture firmware binaries to identify cryptographic primitives and protocols using AI and machine learning.

H2TS Hackathon 2025 - 2nd Place Dec 2025

- Worked with a cross-functional team to build and deploy an AI-driven SOS platform for rapid emergency response.

TECHNICAL SKILLS

- Languages:** Python (Intermediate), Embedded C, MATLAB/Simulink (Intermediate)
- Embedded Systems:** Arduino Uno R3, Arduino Nano, 8051 (AT89S52), Sensor Interfacing, I2C/SPI
- Control & Robotics:** PID Control, Closed-Loop Feedback, Motion Control, ROS 2
- AI & Computer Vision:** Google ADK, Gemini API, FastAPI, Prompt Engineering, Image Processing, Object Detection, AI-based Visual Recognition
- Design & EDA:** SolidWorks, KiCAD, Proteus, Canva
- Tools & Systems:** Linux (Ubuntu) - ROS 2 environment, Binwalk, Firmware Binary Analysis (working knowledge of tools and workflow)

CERTIFICATIONS

- Automation in Manufacturing - IIT Guwahati, NPTEL (Elite, 67%)
- Foundations of Prompt Engineering - AWS Training and Certifications

LEADERSHIP & ACTIVITIES

- Placement Representative (2026-Present) - Coordinate communication between students and placement coordinators
- Joint Treasurer, MIT Quill (2025-26) - Maintained financial records and managed treasury for the college literary club
- Office Bearer & Content Writer, SAHA (2025-26) - Planned and organized academic events for the Science and Humanities Association
- Member, ARME - Association of Robotics and Mechatronics Engineers (2024-Present)
- YRC Volunteer, MIT (2024-Present) - Organized Youth Fest events for visually challenged individuals

LANGUAGES

English (Proficient) | Tamil (Proficient) | Telugu (Conversational) | Hindi (Elementary)